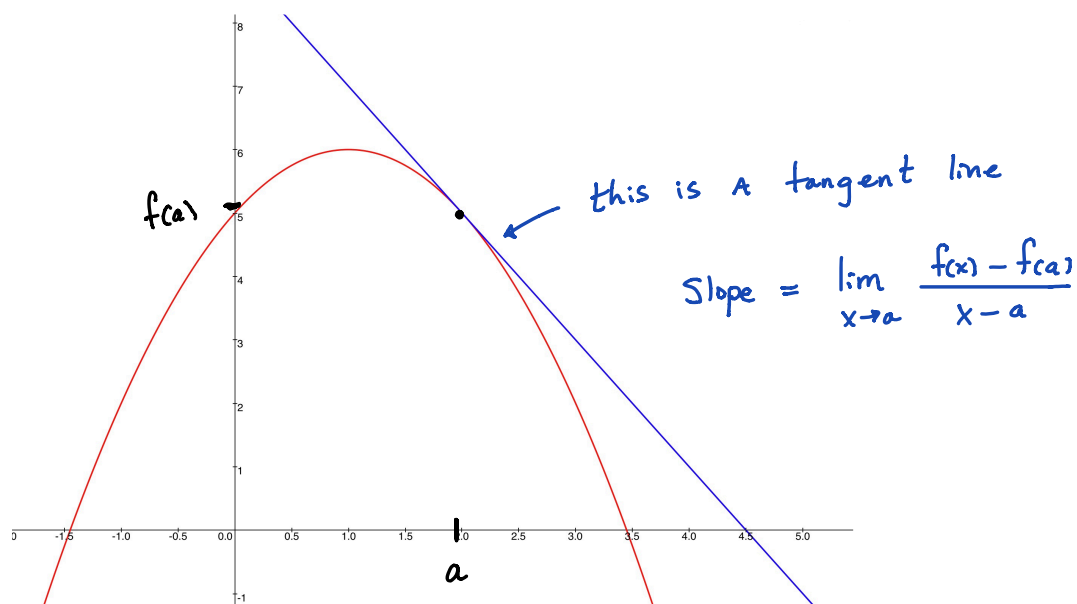
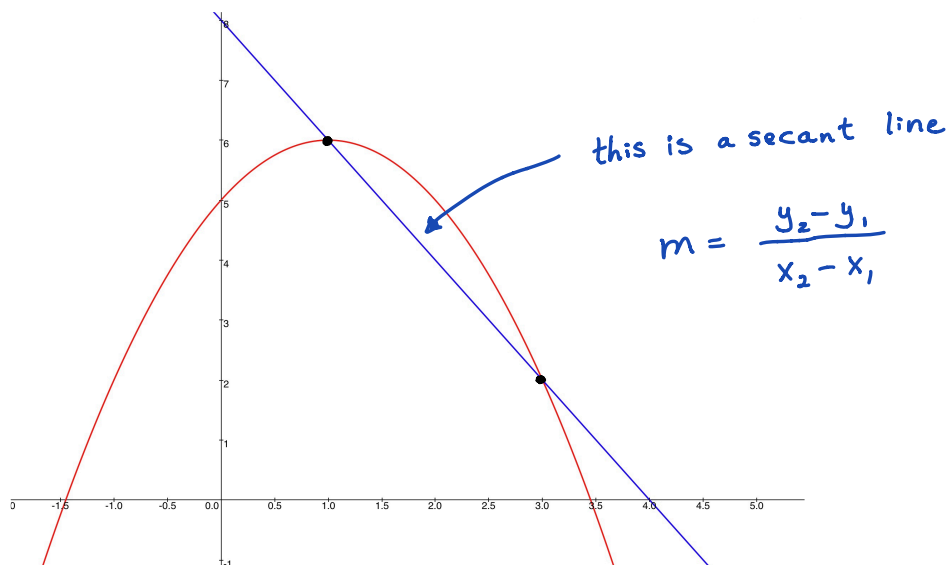


Section 2.7 Derivatives are Rates of Change



Def The derivative of $f(x)$ at a is

$$\lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$$

Ex ① $f(x) = -x^2 + 5x + 2$

A) Find the average rate of change over $[3, 4]$.

↑
slope of the
secant line

$$f(4) = -4^2 + 5(4) + 2 = 6$$

$$f(3) = -3^2 + 5(3) + 2 = 8$$

$$m = \frac{f(4) - f(3)}{4 - 3} = \frac{6 - 8}{1} = \textcircled{-2}$$

B) Find the instantaneous rate of change when $a=3$.

↑
find the derivative

$$\lim_{x \rightarrow 3} \frac{f(x) - f(3)}{x - 3} = \lim_{x \rightarrow 3} \frac{-x^2 + 5x + 2 - 8}{x - 3}$$

$$= \lim_{x \rightarrow 3} \frac{-x^2 + 5x - 6}{x - 3} = \lim_{x \rightarrow 3} \frac{(x-3)(-x+2)}{x-3}$$

$$= \lim_{x \rightarrow 3} (-x+2) = -3+2 = \textcircled{-1}$$

C) Find the equation of the tangent line at $(3, 8)$

From part b, $m = -1$.

use $y - y_1 = m(x - x_1)$

$$y - 8 = -1 \cdot (x - 3)$$

$$y - 8 = -x + 3$$

$$y = -x + 11$$

Ex ② Given $f(x) = \sqrt{x}$, find the equation of the tangent line when $x = 16$.

$$\text{Slope} = \lim_{x \rightarrow 16} \frac{f(x) - f(16)}{x - 16}$$

$$= \lim_{x \rightarrow 16} \frac{\sqrt{x} - \sqrt{16}}{x - 16}$$

$$= \lim_{x \rightarrow 16} \frac{(\sqrt{x} - 4)(\sqrt{x} + 4)}{x - 16} \frac{(\sqrt{x} + 4)}{(\sqrt{x} + 4)}$$

$$= \lim_{x \rightarrow 16} \frac{x - 16}{(x - 16)(\sqrt{x} + 4)}$$

$$= \lim_{x \rightarrow 16} \frac{1}{\sqrt{x} + 4} = \frac{1}{\sqrt{16} + 4} = \frac{1}{8} \text{ slope}$$

Use $m = \frac{1}{8}$ At the point $(16, 4)$

$$y - 4 = \frac{1}{8}(x - 16)$$

$$y = \frac{1}{8}x + 2$$

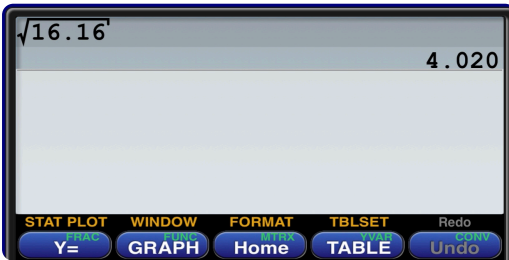
Note This tangent line is a good estimate for $f(x) = \sqrt{x}$ as long as x is close to 16.

Use the tangent line to estimate $\sqrt{16.16}$

we can even
do this without
a calculator! →

$$y = \frac{1}{8}(16.16) + 2$$

$$y = 2.02 + 2 = \textcircled{4.02}$$



← our estimate is spot on!

Ex ③ Suppose you dropped a penny off of the Empire State Building, which is 1250 feet tall.

The height of the penny would be

$$h(t) = -16t^2 + 1250 \text{ After } t \text{ seconds.}$$

A) Find the Average velocity over $[4, 5]$

↑
slope of secant line

$$h(5) = -16(5)^2 + 1250 = 850 \text{ feet}$$

$$h(4) = -16(4)^2 + 1250 = 994 \text{ feet}$$

$$\text{Average velocity} = \frac{850 - 994}{5 - 4} \frac{\text{feet}}{\text{sec}}$$

$$= -144 \frac{\text{ft}}{\text{sec}}$$

B) Find the instantaneous velocity when $t = 7 \text{ sec's}$
↑
slope of tangent line

$$\lim_{t \rightarrow 7} \frac{h(t) - h(7)}{t - 7} = \lim_{t \rightarrow 7} \frac{-16t^2 + 1250 - (-16(7)^2 + 1250)}{t - 7}$$

$$= \lim_{t \rightarrow 7} \frac{-16t^2 + 784}{t - 7}$$

$$= \lim_{t \rightarrow 7} \frac{-16(t^2 - 49)}{t - 7}$$

$$= \lim_{t \rightarrow 7} \frac{-16(t - 7)(t + 7)}{t - 7}$$

$$= \lim_{t \rightarrow 7} -16(t + 7)$$

$$= -16(7 + 7)$$

$$= -224 \frac{\text{ft}}{\text{sec}}$$